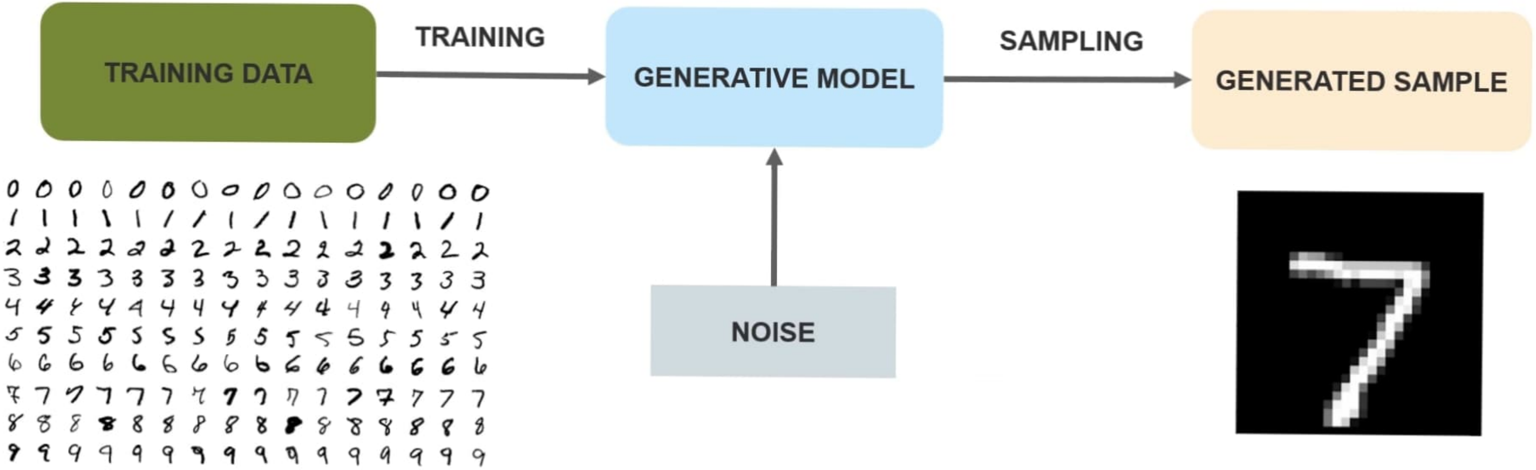
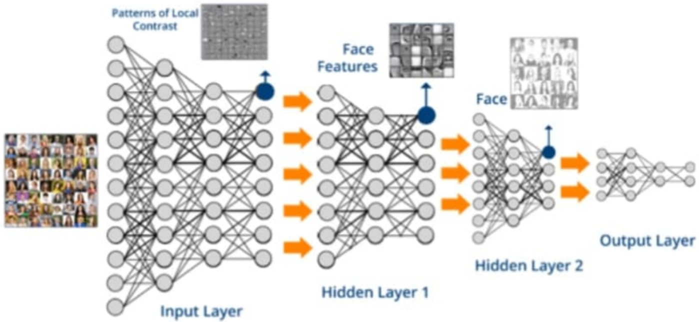
What Are Generative Model?



What Are GANs?



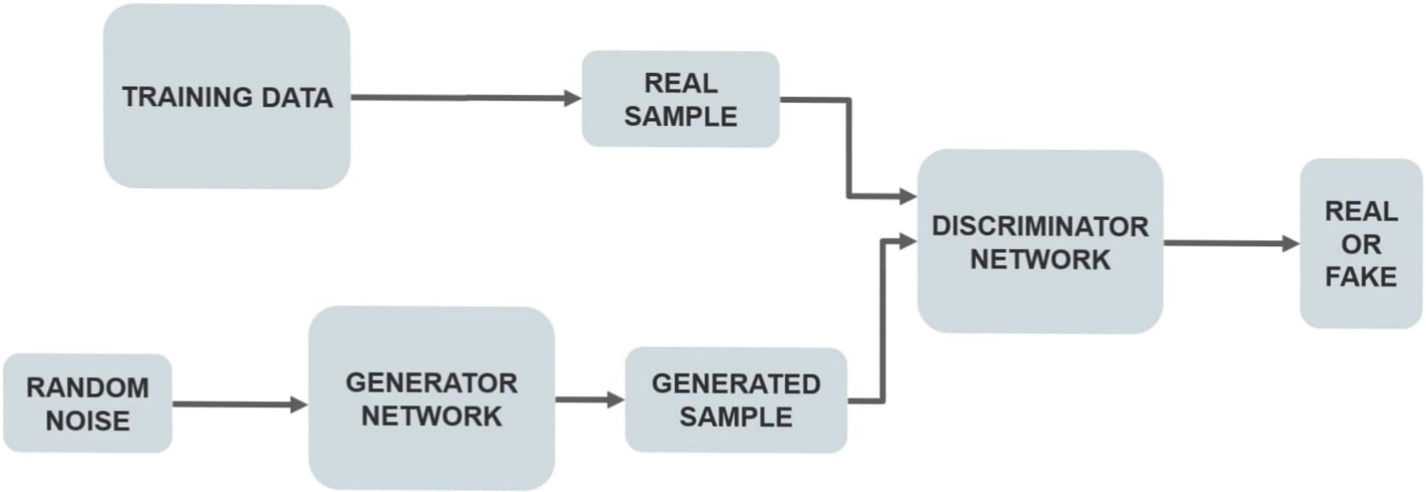
GENERATOR

Generator network that takes a sample and generates a sample of data

DISCRIMINATOR

Discriminator network decides whether the data is generated or taken from the real sample using a binary classification pr oblem with the help of a sigmoid function that gives the output in the range 0 to 1.

How Does A GAN Work?



How Does A GAN Work?

$$V(D,G) = E_{x \sim Pdata(x)} [\log D(x)] + E_{z \sim pz(z)} [\log(1 - D(G(z))]$$

G = Generator	x = sample from real data
D = Discriminator	z = sample from generator
Pdata(x) = Distribution of real data	D(x) = Discriminator Network
Pdata(z) = Distributor of generator	G(z) = Generator Network

How To Train A GAN?

1. Define The Problem
2. Choose Architecture Of GAN
3. Train Discriminator On Real Data
4. Generate Fake Inputs For Generator
5. Train Discriminator On Fake Data
6. Train Generator With The Output Of Discriminator

Challenges Faced By GANs

Problem of stability between generator and discriminator

Problem to determine positioning of the objects

Problem in understanding the perspective

Problem in understanding global objects

GANs Applications

Prediction of Next Frame In A Video

GANs Applications

Text To Image Generation

A Flower With Red Petals And Green Leaves

GANs Applications

Image To Image Translation

Real

Generated

Reconstructed

GANs Applications

Enhancing the Resolution Of An Image